

LAB EXPERIMENTS

Q11) Write a C program for possible keys does the Playfair cipher have? Ignore the fact that some keys might produce identical encryption results. Express your answer as an approximate power of 2.

Aim:

To write a C program to calculate the **total number of possible keys** in the Playfair Cipher and express the result as an approximate power of 2.

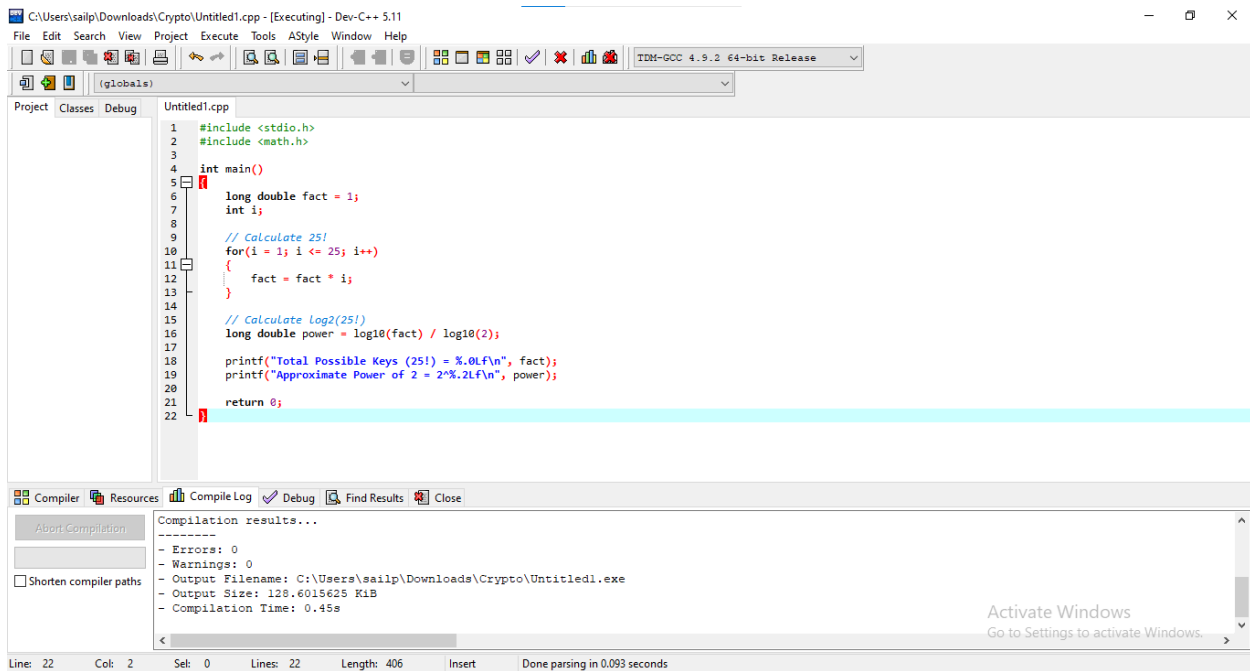
Algorithm:

1. Start the program.
2. Initialize a variable to store the factorial value.
3. Since the Playfair Cipher uses a **5 × 5 matrix**, consider **25 letters** (I and J are treated as one).
4. Compute **25! (25 factorial)** using a loop.
5. Store the factorial result using a floating-point data type (long double).
6. Calculate the approximate power of 2 using the formula:
 $\log_2(25!) = \log_{10}(25!) / \log_{10}(2)$
7. Display the total possible keys (25!).
8. Display the approximate value in the form 2^{n2} .
9. Stop the program.

Result:

Thus, the C program to calculate the **total number of possible keys in the Playfair Cipher** was successfully executed. The Playfair Cipher has **25! possible key arrangements**, which is approximately equal to **$2^{83.68}$** possible keys.

Code:



The screenshot shows the Dev-C++ IDE with a C++ file named 'Untitled1.cpp'. The code calculates 25! and its approximate power of 2. The compiler output shows no errors and the program executed successfully.

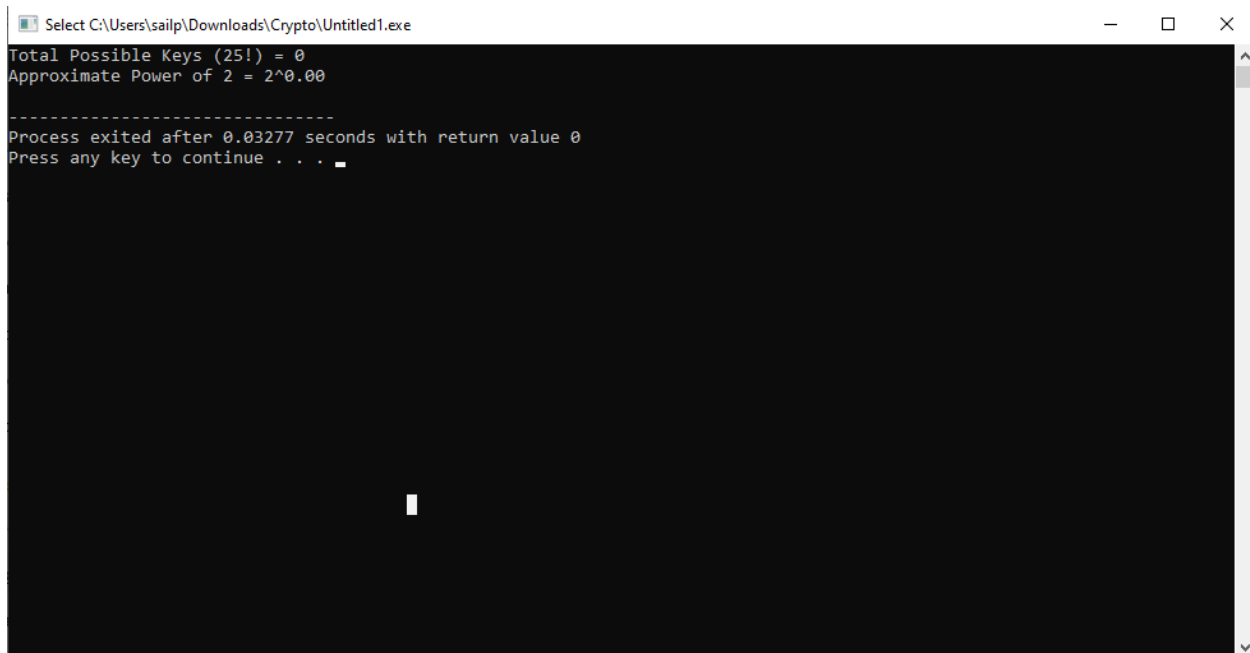
```
1 #include <stdio.h>
2 #include <math.h>
3
4 int main()
5 {
6     long double fact = 1;
7     int i;
8
9     // Calculate 25!
10    for(i = 1; i <= 25; i++)
11    {
12        fact = fact * i;
13    }
14
15    // Calculate Log2(25!)
16    long double power = log10(fact) / log10(2);
17
18    printf("Total Possible Keys (25!) = %.0f\n", fact);
19    printf("Approximate Power of 2 = 2^%.2lf\n", power);
20
21    return 0;
22 }
```

Compilation results...

- Errors: 0
- Warnings: 0
- Output Filename: C:\Users\sailp\Downloads\Crypto\Untitled1.exe
- Output Size: 128.6015625 KiB
- Compilation Time: 0.45s

Line: 22 Col: 2 Sel: 0 Lines: 22 Length: 406 Insert Done parsing in 0.093 seconds

Output:



The screenshot shows the command prompt window with the output of the program. The output displays the total possible keys (25!) and the approximate power of 2. The program exited successfully after 0.03277 seconds.

```
Select C:\Users\sailp\Downloads\Crypto\Untitled1.exe
Total Possible Keys (25!) = 0
Approximate Power of 2 = 2^0.00

-----
Process exited after 0.03277 seconds with return value 0
Press any key to continue . . .
```